

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(c)		Rearrangement for M i.e. $M = \frac{v^2 r}{G}$ (1) Conversion of distance i.e. $20\,000 \times 9.46 \times 10^{15} = (1.892 \times 10^{20})$ (1) Answer = 7.09×10^{39} k[g] (1)		3		3	3	
	(d)		Speed estimated as 110 - 130 [km s ⁻¹] (1) Use of the Doppler equation ecf (1) Answer = 7.7×10^{-5} to 9.1×10^{-5} [m] (1)	1	1 1		3	2	
	(e)		Use of Hubble equation $v = H_0 d$ (1) Answer = 5.0×10^{22} to 5.9×10^{22} [m] (ecf on v) (1)	1	1		2	2	
			Question 2 total	8	6	0	14	9	0